Stellar Rhythms: Probing the Hidden Beat of Stars

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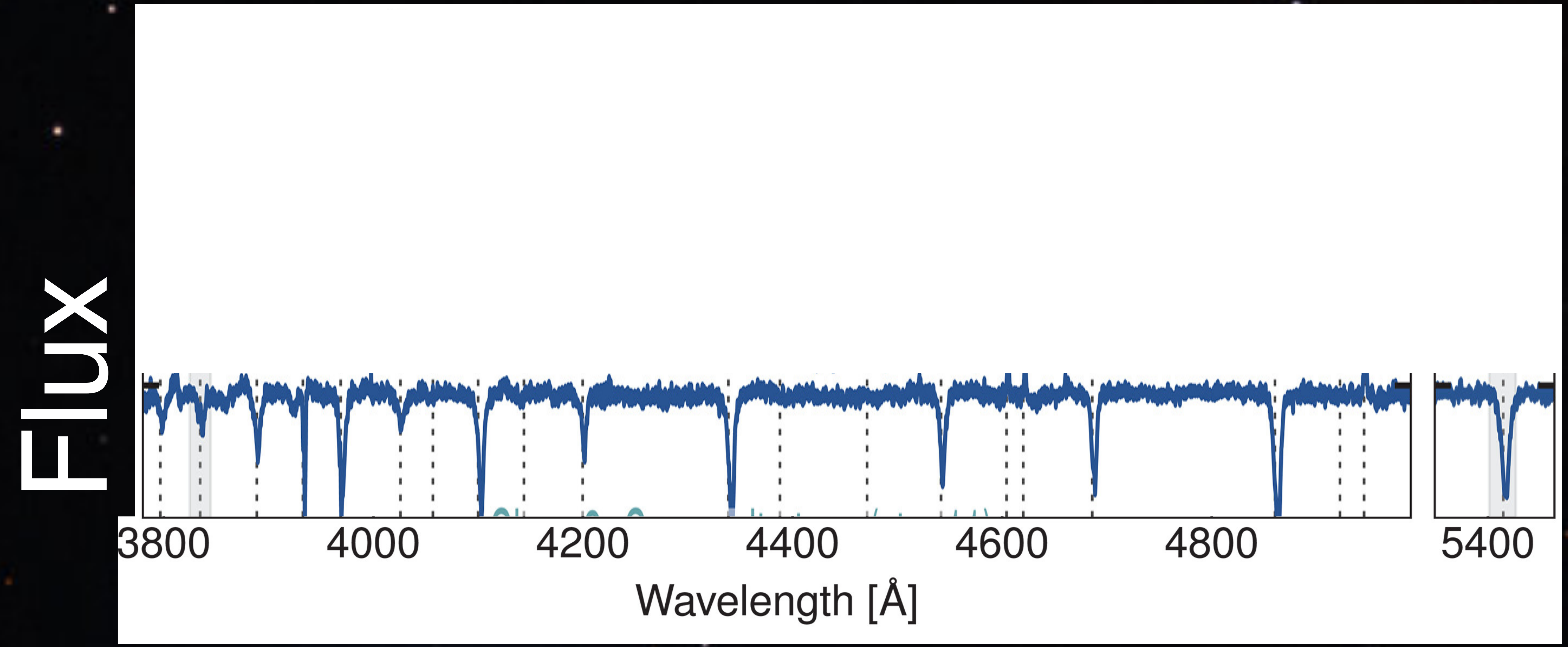
Sun Meng (NAO CAS)


1.Color
2.Brightness
Martin Pugh


1.Color(ν)
2.Brightness
Martin Pugh



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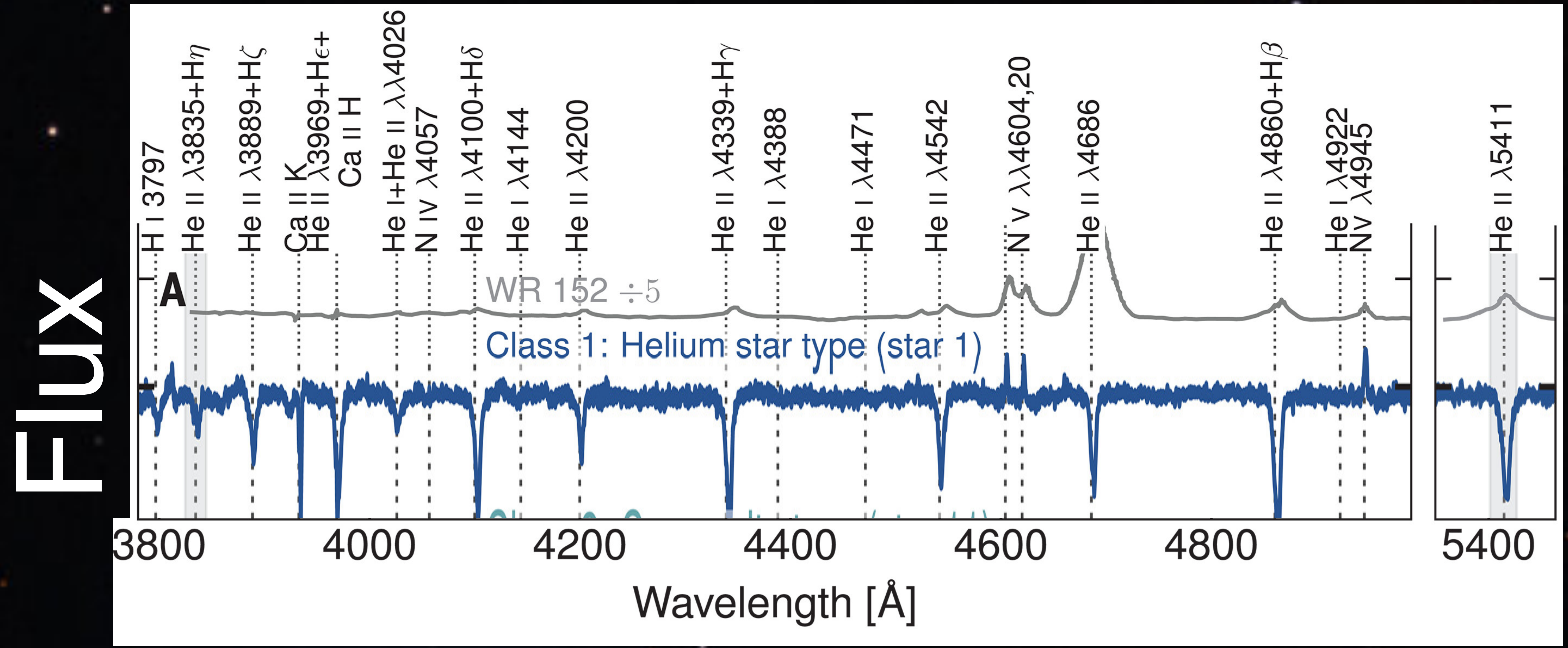

Drout et al 2023

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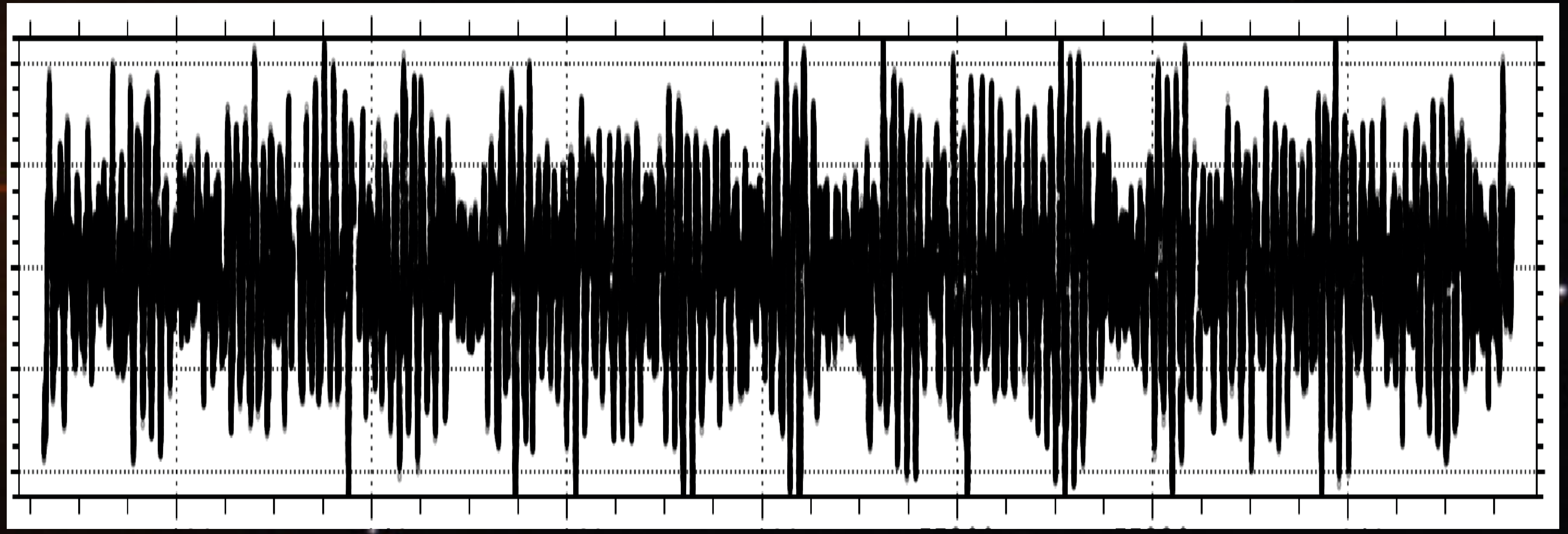
Buysschaert et al 2018

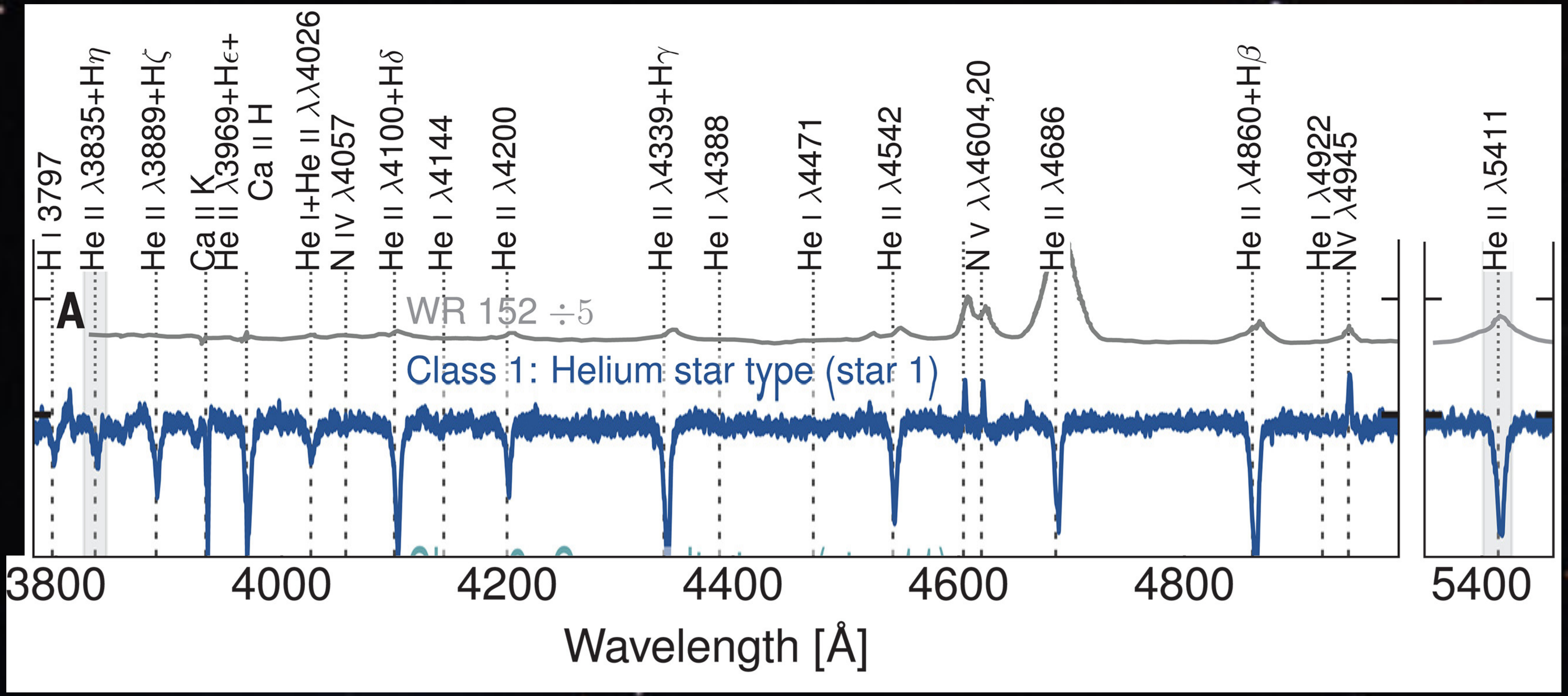
HD 43317

Flux



time

Flux



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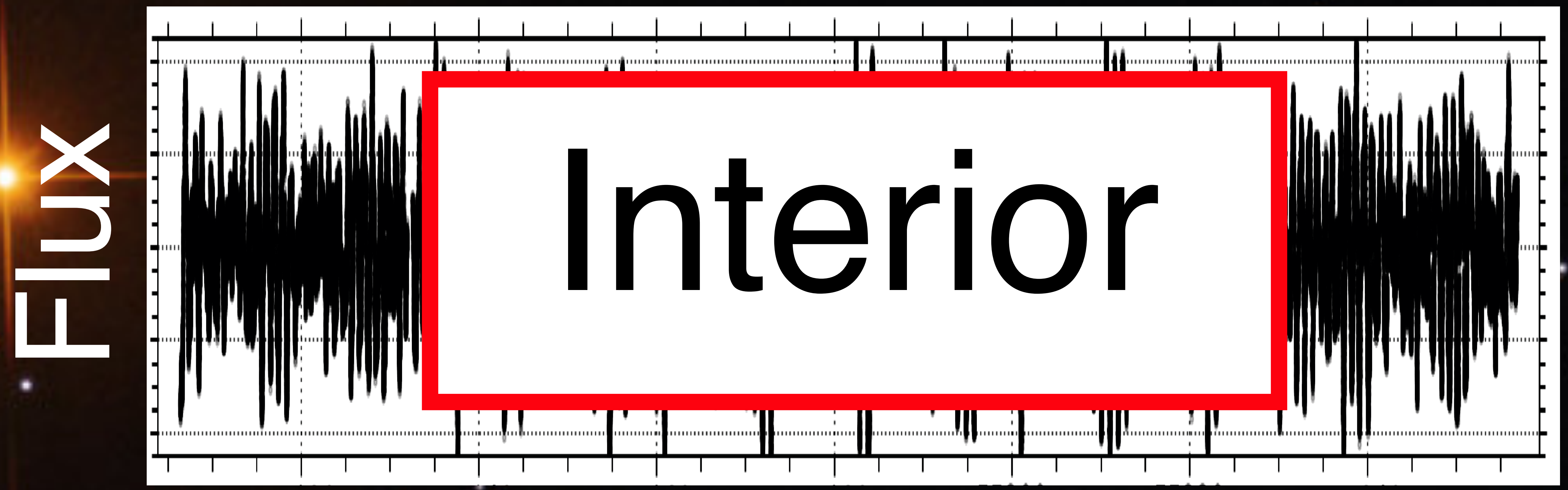

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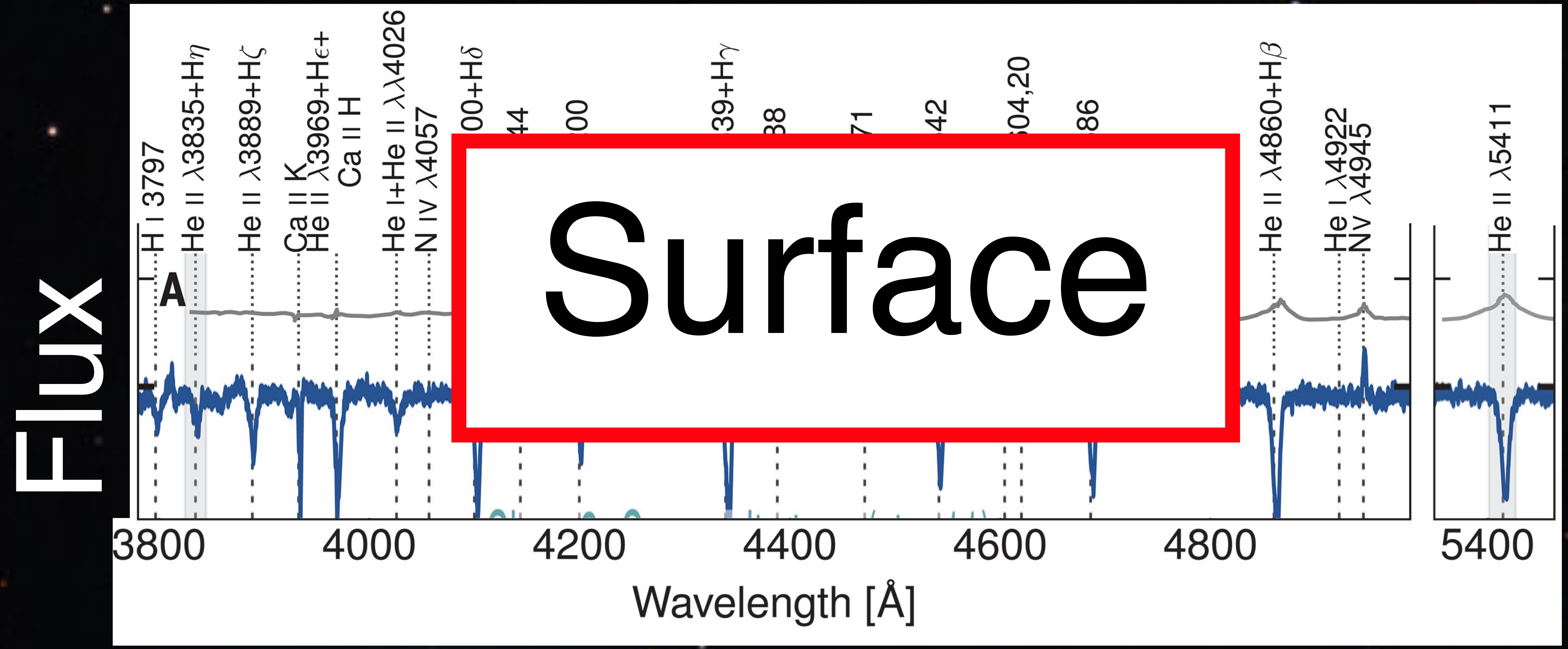

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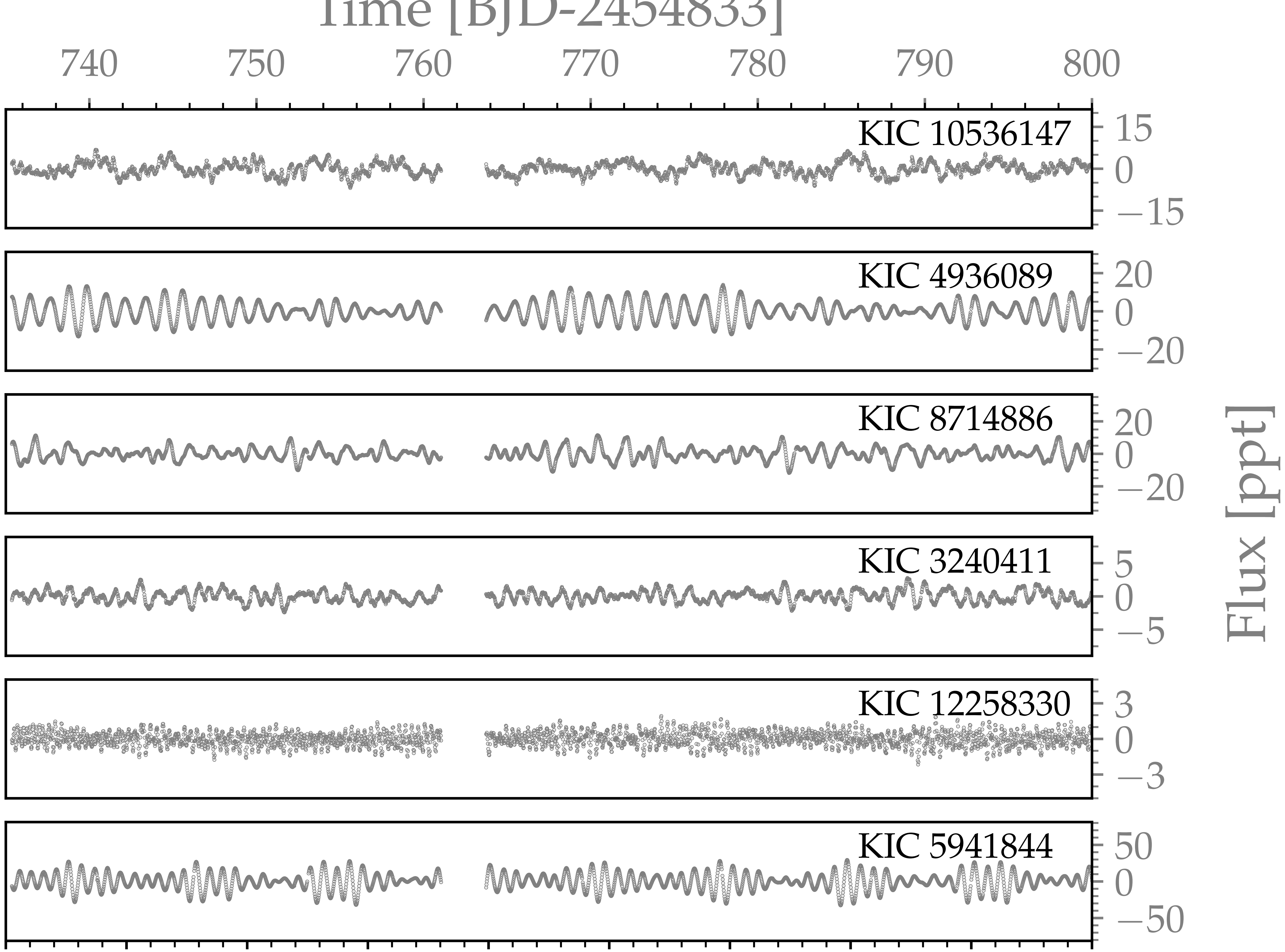
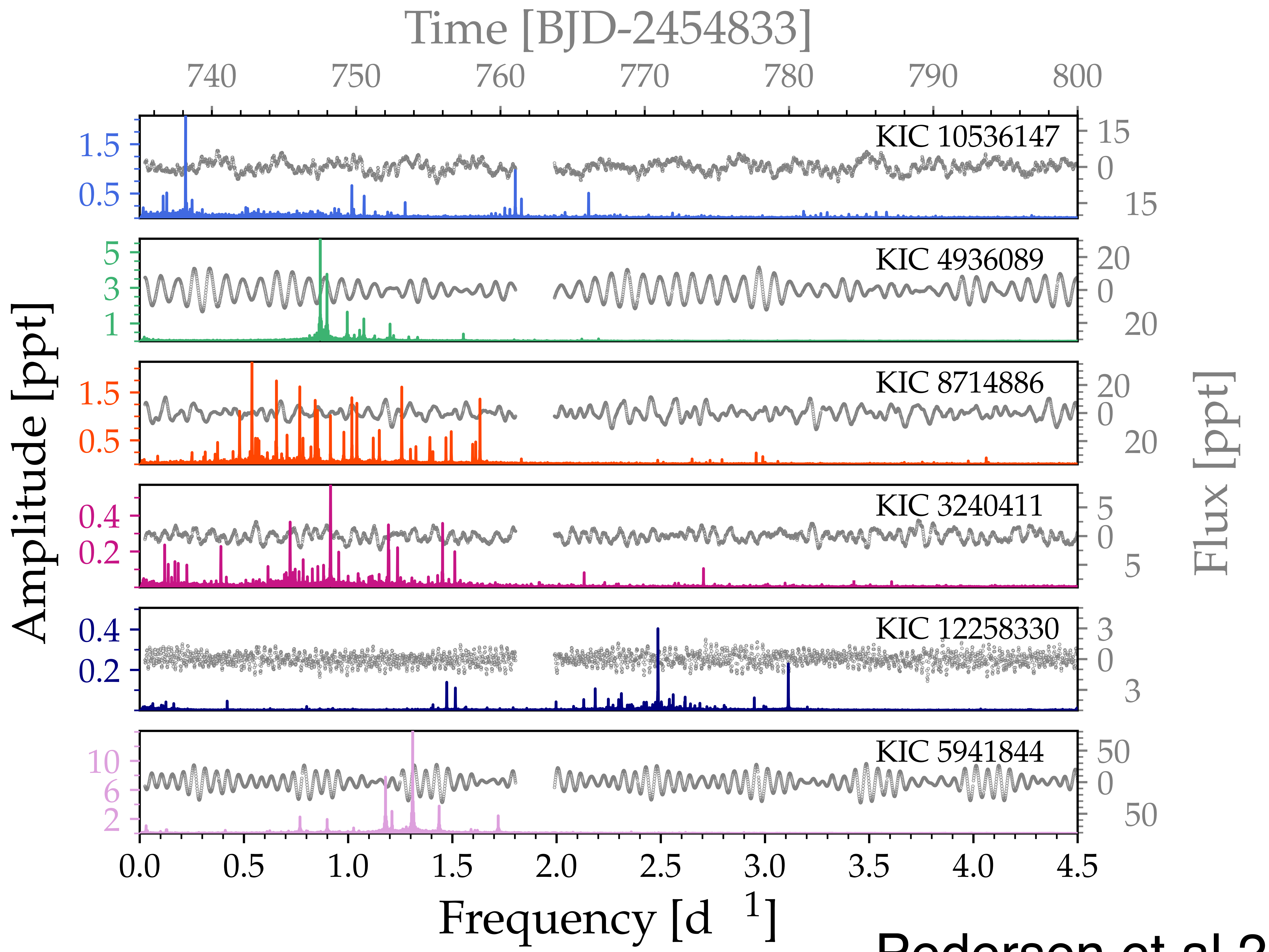


Pedersen et al 2021

Asteroseismology



Pedersen et al 2021

Fluid Dynamics

Fluid Dynamics

$$\rho \frac{\partial \boldsymbol{u}}{\partial t} + \rho \boldsymbol{u} \cdot \nabla \boldsymbol{u} + \nabla p = \rho \boldsymbol{g} + \nu \nabla^2 \boldsymbol{u}$$

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+ Equation of State

$$T = T(\rho, s) \qquad p = p(\rho, s)$$

Oscillations

$$\rho \frac{\partial \boldsymbol{u}}{\partial t} + \rho \boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} + \boldsymbol{\nabla} p = \rho \boldsymbol{g} + \nu \nabla^2 \boldsymbol{u}$$

Linearize:

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$$\boldsymbol{u} \to \boldsymbol{u}'$$

$$\rho \to \rho_0 + \rho'$$

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Coefficients only a function of *r*

-> spin-weighted spherical harmonics, exp. in time

Spherical Harmonics

Vasil et al 2019

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$$\rho'(r, \theta, \phi, t) = \sum_{\ell, m, n} \rho'_{\ell, m, n}(r) Y^0_{\ell, m}(\theta, \phi) \exp(-i \omega_n t)$$

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σ=-1,0,1

$$\boldsymbol{u}'(r, \theta, \phi, t) = \sum_{\ell, m, n, \sigma} \boldsymbol{u}'_{\ell, m, n, \sigma}(r) Y^\sigma_{\ell, m}(\theta, \phi) \boldsymbol{e}_\sigma \exp(-i \omega_n t)$$

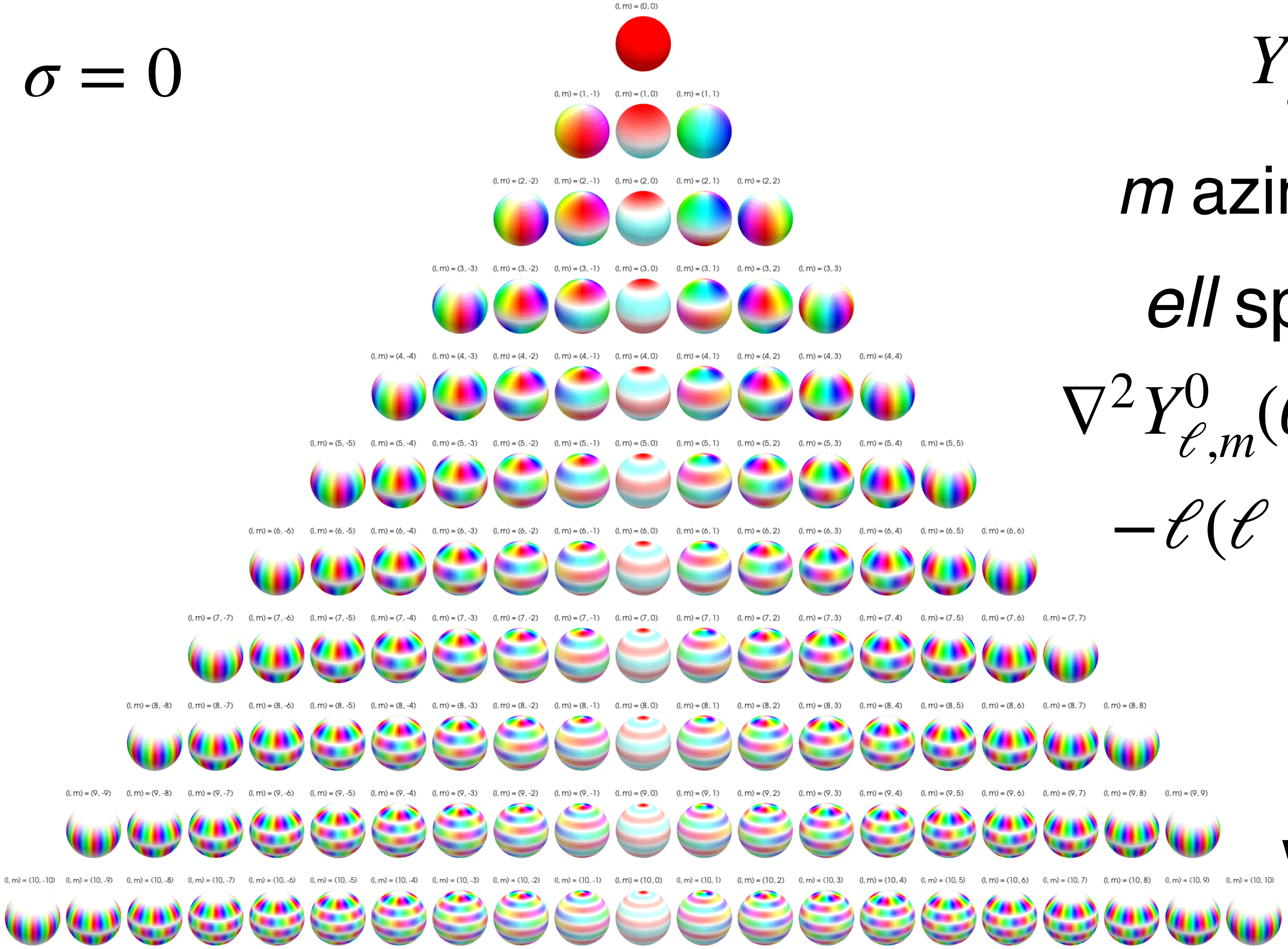
$$\sigma = 0$$

$$Y^0_{\ell,m}(\theta, \phi)$$

m azimuthal order

ell spatial scale

$$\nabla^2 Y^0_{\ell,m}(\theta, \phi) =$$
$$-\ell(\ell + 1) Y^0_{\ell,m}(\theta, \phi)$$



wikipedia

Spherical Harmonics

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Vasil et al 2019

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Angular derivative only are functions of *ell* not *m*!!!

Eigenvalue Problem

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$$X'_{\ell,m,n} = \begin{bmatrix} \boldsymbol{u}'_{\ell,m,n}(r) \\ \rho'_{\ell,m,n}(r) \\ s'_{\ell,m,n}(r) \end{bmatrix}$$

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$$L\left(\ell, \omega_n, r, \frac{d}{dr}\right) . X'_{\ell,m,n} = 0$$

Eigenvalue Problem

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ω_n is *eigenvalue*

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Numerical solver (GYRE)

Approximation -> analytic solution

Oscillation Equation

$$\rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} + \nabla p' = \rho' \boldsymbol{g}_0 + \rho_0 \boldsymbol{g}' + \nu \nabla^2 \boldsymbol{u}'$$

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$$\boxed{\frac{d^2 \Upsilon'(r)}{dr^2} + \left(\frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell + 1)}{r^2} \left(\frac{N^2}{\omega_n^2} - 1 \right) \right) \Upsilon'(r) = 0}$$

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sound speed

Oscillation Equation

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$$c_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s$$

sound speed

$$N^2 = \frac{g}{c_p}\frac{ds}{dr}$$

Brunt-Vaisala frequency

Oscillation Equation

$$\frac{d^2\Upsilon'(r)}{dr^2} + \left(\frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)\right)\Upsilon'(r) = 0$$

$$\Upsilon'_{\ell,m,n} = \rho_0^{1/2} c_s^2 \nabla \cdot \boldsymbol{u}'_{\ell,m,n}$$

$$c_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s$$

sound speed

$$N^2 = \frac{g}{c_p}\frac{ds}{dr}$$

Brunt-Vaisala frequency

Acoustic cutoff-frequency

$$\omega_c^2 = \frac{c_s^2}{2}\frac{\rho_0''}{\rho_0} - \frac{c_s^2}{4}\left(\frac{\rho_0'}{\rho_0}\right)^2 + \frac{\rho_0'}{\rho_0}c_s^{2'} + c_s^{2''}$$

Asymptotic Modes

$$\frac{d^2\Upsilon'(r)}{dr^2} + \left(\frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)\right)\Upsilon'(r) = 0$$

$$\Upsilon'_{\ell,m,n} = \rho_0^{1/2} c_s^2 \nabla \cdot \boldsymbol{u}'_{\ell,m,n}$$

WKB approximation

Asymptotic Modes

$$\frac{d^2\Upsilon'(r)}{dr^2} + \left(\frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)\right)\Upsilon'(r) = 0$$

$$\Upsilon'_{\ell,m,n} = \rho_0^{1/2} c_s^2 \nabla \cdot \boldsymbol{u}'_{\ell,m,n}$$

WKB $\Upsilon'_{\ell,m,n}(r) \sim \exp(ik_{r,n}r)$

$$\frac{d\Upsilon'_{\ell,m,n}(r)}{dr} = ik_{r,n}\Upsilon'_{\ell,m,n}(r)$$

Asymptotic Modes

$$\frac{d^2\Upsilon'(r)}{dr^2} + \left(\frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)\right)\Upsilon'(r) = 0$$

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WKB $\Upsilon'_{\ell,m,n}(r) \sim \exp(ik_{r,n}r)$

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$$k_{r,n}^2 = \frac{\omega^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega^2} - 1\right)$$

Asymptotic Modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell + 1)}{r^2} \left(\frac{N^2}{\omega_n^2} - 1 \right)$$

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \gg N^2, \omega_c^2$

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \gg N^2, \omega_c^2$

$$k_{r,n}^2 = \frac{\omega_n^2}{c_s^2} - \frac{\ell(\ell+1)}{r^2}$$

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \gg N^2, \omega_c^2$

$$S_\ell^2 = \frac{\ell(\ell+1)c_s^2}{r^2}$$

$$k_{r,n}^2 = \frac{\omega_n^2}{c_s^2} - \frac{\ell(\ell+1)}{r^2} = \frac{\omega_n^2 - S_\ell^2}{c_s^2}$$

Lamb freq.

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

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Lamb freq.

Lamb (1895)

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

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Lamb freq.

Lamb (1895)

$$k_{r,n}^2 = \frac{\omega_n^2}{c_s^2} - \frac{\ell(\ell+1)}{r^2} = \frac{\omega_n^2 - S_\ell^2}{c_s^2}$$

$$\omega_n^2 = c_s^2 k_{r,n}^2 + S_\ell^2$$

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

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$$S_\ell^2 = \frac{\ell(\ell+1)c_s^2}{r^2}$$

Lamb freq.

Boundaries at r=0, r=R

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \gg N^2, \omega_c^2$

$$S_\ell^2 = \frac{\ell(\ell+1)c_s^2}{r^2}$$

$$\omega_n^2 = c_s^2 k_{r,n}^2 + S_\ell^2$$

$$\lambda_{r,n} = \frac{R_\star}{n}$$

Lamb freq.

Boundaries at r=0, r=R

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \gg N^2, \omega_c^2$

$$\omega_n^2 = c_s^2 k_{r,n}^2 + S_\ell^2$$

$$S_\ell^2 = \frac{\ell(\ell+1)c_s^2}{r^2}$$

Lamb freq.

Boundaries at r=0, r=R

$$\lambda_{r,n} = \frac{R_\star}{n}$$

$$k_{r,n} = \frac{2\pi}{\lambda_{r,n}} = \frac{2\pi n}{R_\star}$$

Asymptotic p-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \gg N^2, \omega_c^2$

$$S_\ell^2 = \frac{\ell(\ell+1)c_s^2}{r^2}$$

Lamb freq.

$$\omega_n^2 = c_s^2 k_{r,n}^2 + S_\ell^2$$

$$\lambda_{r,n} = \frac{R_\star}{n}$$

Boundaries at r=0, r=R

$$k_{r,n} = \frac{2\pi}{\lambda_{r,n}} = \frac{2\pi n}{R_\star} \quad \omega_n = \frac{2\pi c_s}{R_\star} n$$

Asymptotic g-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell + 1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \ll N^2, \omega_c^2$

Asymptotic g-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \ll N^2, \omega_c^2$

$$k_{r,n}^2 = \frac{\ell(\ell+1)}{r^2}\frac{N^2}{\omega_n^2}$$

Asymptotic g-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

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$$k_{r,n}^2 = \frac{\ell(\ell+1)}{r^2}\frac{N^2}{\omega_n^2}$$

$$\omega_n^2 = \frac{\ell(\ell+1)}{r^2}\frac{N^2}{k_{r,n}^2}$$

Asymptotic g-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

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$$k_{r,n}^2 = \frac{\ell(\ell+1)}{r^2}\frac{N^2}{\omega_n^2}$$

$$\omega_n^2 = \frac{\ell(\ell+1)}{r^2}\frac{N^2}{k_{r,n}^2}$$

Boundaries at r=0, r=R

$$k_{r,n} = \frac{2\pi n}{R_\star}$$

Asymptotic g-modes

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

Limits: $\omega_n^2 \ll N^2, \omega_c^2$

$$k_{r,n}^2 = \frac{\ell(\ell+1)}{r^2}\frac{N^2}{\omega_n^2}$$

$$\omega_n^2 = \frac{\ell(\ell+1)}{r^2}\frac{N^2}{k_{r,n}^2}$$

Boundaries at r=0, r=R

$$k_{r,n} = \frac{2\pi n}{R_\star}$$

$$\omega_n = \frac{\sqrt{\ell(\ell+1)}}{r}\frac{NR_\star}{2\pi n}$$

Asymptotic Oscillations

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

p-modes

$$\omega_n = \frac{2\pi c_s}{R_\star} n$$

g-modes

$$\omega_n = \frac{\sqrt{\ell(\ell+1)}}{r}\frac{NR_\star}{2\pi n}$$

Asymptotic Oscillations

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

p-modes

$$\omega_n = \frac{2\pi c_s}{R_\star} n$$

high n, high ω

$$n_{pg} > 0$$

g-modes

$$\omega_n = \frac{\sqrt{\ell(\ell+1)}}{r}\frac{NR_\star}{2\pi n}$$

high n, low ω

$$n_{pg} < 0$$

Asymptotic Oscillations

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

p-modes

$$\omega_n = \frac{2\pi c_s}{R_\star} n$$

$$\omega_{n+1} - \omega_n = \Delta\omega$$

g-modes

$$\omega_n = \frac{\sqrt{\ell(\ell+1)}}{r}\frac{NR_\star}{2\pi n}$$

Asymptotic Oscillations

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

p-modes

$$\omega_n = \frac{2\pi c_s}{R_\star} n$$

$$\omega_{n+1} - \omega_n = \Delta\omega$$

$$\Delta\omega = \frac{2\pi c_s}{R_\star}$$

g-modes

$$\omega_n = \frac{\sqrt{\ell(\ell+1)}}{r}\frac{NR_\star}{2\pi n}$$

Asymptotic Oscillations

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

p-modes

$$\omega_n = \frac{2\pi c_s}{R_\star} n$$

$$\omega_{n+1} - \omega_n = \Delta\omega$$

$$\Delta\omega = \frac{2\pi c_s}{R_\star}$$

g-modes

$$\omega_n = \frac{\sqrt{\ell(\ell+1)}}{r}\frac{NR_\star}{2\pi n}$$

$$P_n = \frac{2\pi}{\omega_n} = \frac{nr}{NR_\star\sqrt{\ell(\ell+1)}}$$

Asymptotic Oscillations

$$k_{r,n}^2 = \frac{\omega_n^2 - \omega_c^2}{c_s^2} + \frac{\ell(\ell+1)}{r^2}\left(\frac{N^2}{\omega_n^2} - 1\right)$$

p-modes

$$\omega_n = \frac{2\pi c_s}{R_\star} n$$

$$\omega_{n+1} - \omega_n = \Delta\omega$$

$$\Delta\omega = \frac{2\pi c_s}{R_\star}$$

g-modes

$$\omega_n = \frac{\sqrt{\ell(\ell+1)}}{r} \frac{N R_\star}{2\pi n}$$

$$P_{n+1} - P_n = \Delta P = \frac{r}{N R_\star \sqrt{\ell(\ell+1)}}$$